

3 Animals and plants respond to changes in their environment.

(a) The mammalian eye contains photoreceptor cells which respond to light.

The photographs show a grey-bellied night monkey and a squirrel monkey.



Grey-bellied night monkey



Squirrel monkey

(Source: © rod williams/Alamy Stock Photo) (Source: © Jesse Kraft/Alamy Stock Photo)

The grey-bellied night monkey is active at night and the squirrel monkey is active during the day.

Rod cells are photoreceptors located in the retina. Rod cells detect low light intensities.

Cone cells are also photoreceptors located in the retina. They are responsible for colour vision and require higher light intensities.

The numbers of different photoreceptors in the retina of one eye of each of these two species of monkey were compared.

The results are shown in the table.

Species	Number of photoreceptors in the retina	
	Cone cells	Rod cells
Grey-bellied night monkey	6.0×10^6	1.4×10^8
Squirrel monkey	7.0×10^6	3.5×10^7

- (i) A section of squirrel monkey retinal tissue contained a total of 5 000 photoreceptors.

Calculate how many of these photoreceptors would be rod cells.

Assume an even distribution of rod cells and cone cells in the retina.

(2)

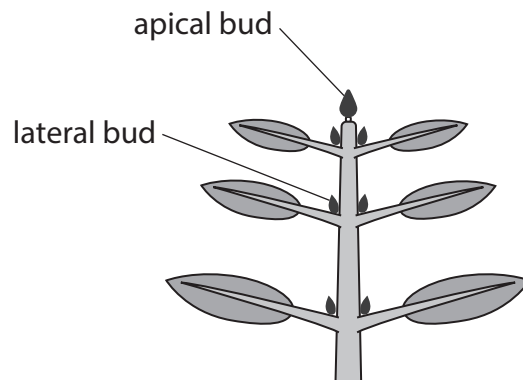
Answer

- (ii) Explain why there is a difference in the numbers of rod cells and cone cells in the retinas of the grey-bellied night monkey and the squirrel monkey.

(3)

(b) Plants also respond to their environment.

The diagram shows a seedling with the apical bud and a lateral bud labelled.



In an investigation, three genetically identical seedlings, with the same shoot length, had their apical buds removed.

The table shows the treatments for each seedling and their appearance after being grown next to a source of light for five days.

The arrows show the direction of the light.

Seedling 1	Seedling 2	Seedling 3
no further treatment	auxin applied to the cut top of the seedling	gibberellins applied to the cut top of the seedling

Deduce the effect of each treatment on seedlings that have had their apical buds removed.

Use the information in the table to support your answer.

(3)

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(Total for Question 3 = 8 marks)